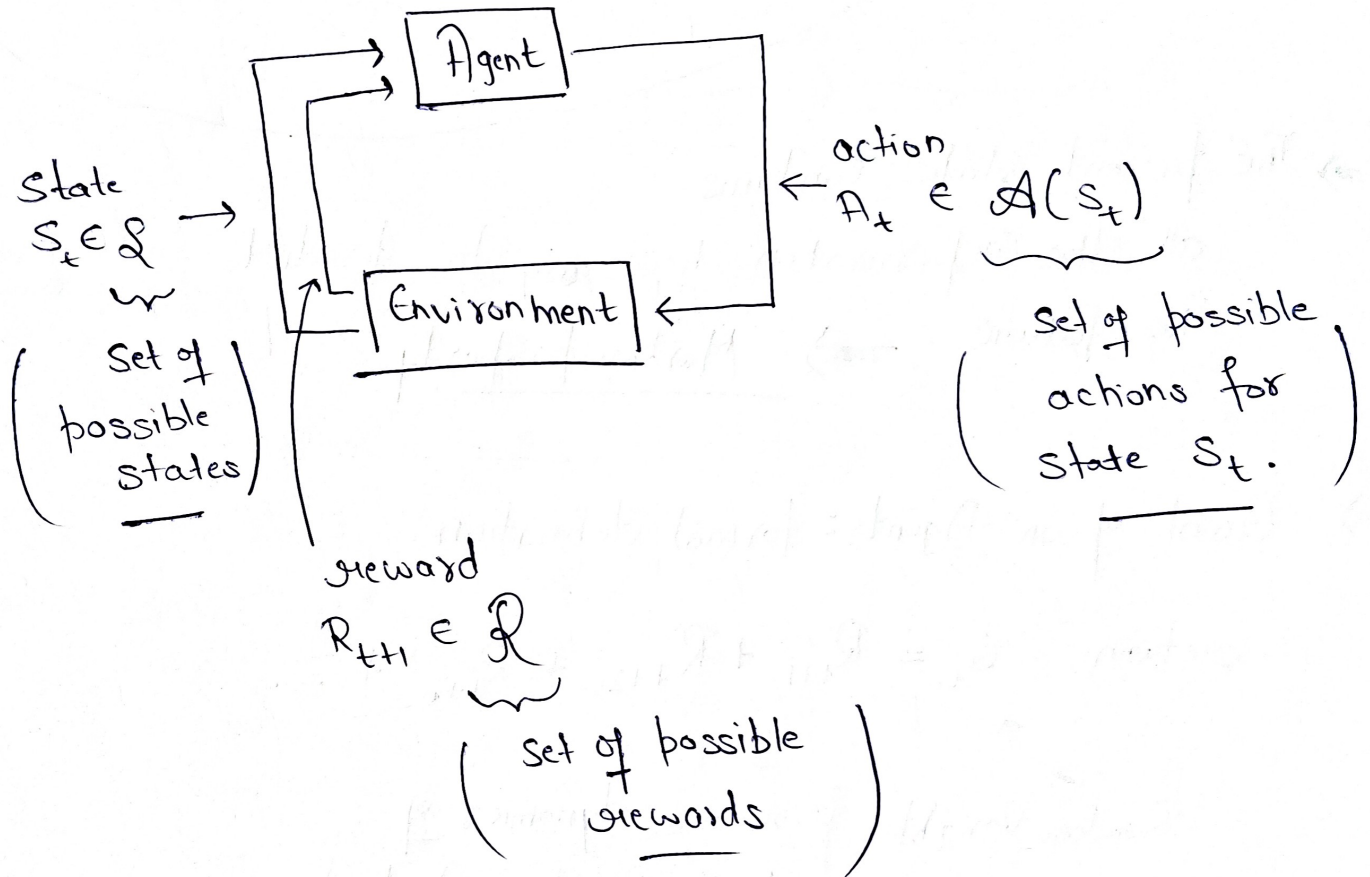
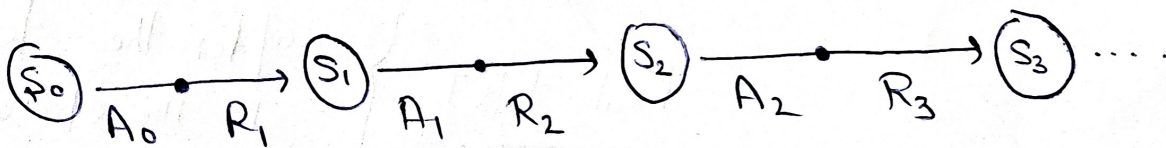


Week 2 →

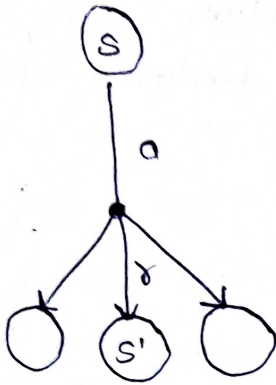
Markov Decision Process →



→ The k -armed bandit problem can not capture
"the actions we choose now affect the amount
of reward we can get into the future."



→ Dynamics of an MDP →



$$p(s', r | s, a)$$

$$p: \mathcal{S} \times \mathcal{R} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$$

$$\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r | s, a) = 1 \quad \forall s \in \mathcal{S}, a \in \mathcal{A}(s)$$

→ The present state contains all the information necessary to predict the future → Markov property

→ Goal of an Agent: formal definition

return $G_t = R_{t+1} + R_{t+2} + R_{t+3} + \dots$

Random Variable } because dynamics of MDP can be stochastic

maximise the expected reward

$$E[G_t] = E[R_{t+1} + R_{t+2} + \dots + R_T]$$

Final step T } after the step T the episode will end!

Continuing and Episodic Tasks →

(6)

Episodic Tasks

- ∅ Interaction breaks naturally into episodes.
- ∅ Each episode ends in a terminal state.
- ∅ episodes are independent.
- ∅ $G_t = R_{t+1} + R_{t+2} + \dots + R_T$

Continuing Tasks

- ∅ Interaction goes on continually.
- ∅ No terminal state.
- ∅ $G_t = R_{t+1} + R_{t+2} + \dots + \dots = \infty?$

How to make this finite?

→ Discounting → Discount the Rewards in the future }
by γ .
where $0 \leq \gamma < 1$

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots + \gamma^{k-1} R_{t+k} + \dots$$

$$\boxed{G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}} \rightarrow \text{Finite!}$$

⇓ why?

∅ Assume $R_{\max}$ is the maximum reward the agent can receive.

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \leq \sum_{k=0}^{\infty} \gamma^k R_{\max}$$

$$= R_{\max} \left(\sum_{k=0}^{\infty} \gamma^k \right) \rightarrow \text{Converging geometric series}$$

$$G_t \leq R_{\max} \times \underbrace{\frac{1}{1-\gamma}}$$

Finite!

⇒ Effect of γ on agent behaviour

$$1) \quad \gamma = 0 \quad \Rightarrow \quad G_t = R_{t+1} \quad \left. \vphantom{\begin{matrix} 1) \end{matrix}} \right\} \text{Short sighted agent}$$

$$2) \quad \gamma \rightarrow 1 \quad \Rightarrow \quad \text{—————} \quad \left. \vphantom{\begin{matrix} 2) \end{matrix}} \right\} \text{far-sighted agent}$$

⇒

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$G_t = R_{t+1} + \gamma \left(R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots \right)$$

$$\boxed{G_t = R_{t+1} + \gamma G_{t+1}} \star$$